LINUX LAB PROJECT REPORT

L47 – Bulk Photo Renamer

A menu-driven Bash shell scripting project

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Project	L47 – Bulk Photo Renamer
Platform	Linux / Ubuntu Terminal

Submitted as a Linux lab project demonstrating shell scripting, file operations, menu-driven programming, logging, backup creation, and task scheduling.

Academic Project Report

1. Introduction

Managing many files manually can take time and may lead to inconsistent filenames. The **Bulk Photo Renamer** project is a menu-driven Bash script designed to help a user preview and rename multiple files, keep a record of rename operations, create backups, and review activity logs from the Linux terminal. It uses common Linux commands and shell programming concepts.

2. Problem Statement

When a folder contains many photos or other files, renaming each item one by one is slow and makes it difficult to follow a consistent naming pattern. The project provides a command-line menu so the user can view proposed names, rename files in batches, retain rename history, and create a compressed backup before or after changes.

3. Objectives

• Create a simple menu-driven program using Bash.

• Preview new filenames before changing existing names.

• Rename files using a prefix, date, and counter.

• Rename files based on their modification date.

• Record rename history to support undo operations.

• View the data folder and recent activity log.

• Create a compressed .tar.gz backup of project data.

• Run an automatic incoming-file task using cron.

4. Tools and Technologies

Tool / Command	Purpose
Bash shell	Writes and runs the menu-driven script and helper functions.
Ubuntu / Linux terminal	Runs commands and provides the project environment.
for, while, case	Loops through files and handles menu choices.
mv, date, printf	Renames files and creates date/counter-based names.
tar -czf	Creates compressed backup archives.
crontab	Schedules the automatic task.
nano, chmod, ls, tail	Edits scripts, sets executable permission, lists files, and views logs.

5. Project Structure

The project is stored in `~/linux_project/photorenamer`. The main script displays the menu, while helper functions are kept in a separate file.

```
photorenamer/
■■■ main.sh Main menu and command-line arguments
■■■ lib/
■ ■■■ functions.sh Helper functions and rename features
■■■ data/
■ ■■■ photos/ Files used for testing
■■■ incoming/ Files waiting for the scheduled task
■■■ backups/ Compressed backup archives
■■■ logs/
■ ■■■ activity.log Record of project actions
■ ■■■ undo.log Saved old/new filename pairs
■■■ reports/ Space for report materials
```

6. Main Features

Option 1 – Preview new names: Shows existing filenames alongside the proposed date/counter-based names without renaming them.

Option 2 – Rename with prefix: Asks for a prefix, then renames files using the prefix, current date, and a counter.

Option 3 – Rename by modified date: Uses each file's modification date when forming its new name.

Option 4 – View undo history: Displays the saved rename history so the user can review recorded changes.

Option 5 – Undo renames: Uses saved old/new filename pairs to attempt to restore previous names.

Option 6 – View data: Lists files in the data folder.

Option 7 – Backup data: Creates a timestamped compressed archive in the backups folder.

Option 8 – View activity log: Displays the latest entries from the activity log.

Cron task: Processes files placed in the incoming folder and moves them into the data folder with a date-based prefix.

7. How to Run the Project

```
cd ~/linux_project/photorenamer
chmod +x main.sh
./main.sh
```

For help, run `./main.sh --help`. To create a backup directly, run `./main.sh --backup`. To test the automatic task manually, run `./main.sh --cron`.

8. Testing and Results

Use the following checklist during the demonstration. Record the actual result after running each option on the Ubuntu system. Insert screenshots from your own terminal in the spaces below; the report does not contain fabricated terminal screenshots.

Test	Expected result	Result
Run ./main.sh	The project menu appears.	To be verified
Option 1	Proposed filenames are displayed; files remain unchanged.	To be verified
Options 2 and 3	Files are renamed using the selected pattern/date.	To be verified
Options 4 and 5	History is shown and undo is attempted using the log.	To be verified
Options 6 and 8	File listing and recent activity are displayed.	To be verified
Option 7 / --backup	A .tar.gz archive is created in backups/.	To be verified
--cron	Files in incoming/ are processed and logged.	To be verified

Screenshot Evidence (add your own screenshots)

Screenshot 1 – Main menu

Paste or insert a screenshot from your Ubuntu terminal here.

Screenshot 2 – Rename / preview

Paste or insert a screenshot from your Ubuntu terminal here.

Screenshot 3 – Backup and logs

Paste or insert a screenshot from your Ubuntu terminal here.

Screenshot 4 – Cron task

Paste or insert a screenshot from your Ubuntu terminal here.

9. Conclusion

The Bulk Photo Renamer project demonstrates how Bash scripting can simplify repetitive file management tasks in Linux. It combines a menu, loops, conditional statements, functions, file operations, logging, compressed backups, and cron scheduling. With careful testing and valid rename history, the script can help users apply consistent naming patterns and review their actions.

10. Learning Outcomes

• Learned to navigate project folders and run scripts in the Linux terminal.

• Practised Bash variables, functions, loops, case statements, and input validation.

• Used commands such as mv, date, printf, ls, tail, and tar.

• Learned how to store activity and rename history in log files.

• Learned the basic purpose of cron for scheduling repeated tasks.

11. Important Commands

Command	Use
`./main.sh`	Starts the interactive menu.
`chmod +x main.sh`	Makes the script executable.
`bash -n main.sh`	Checks the script for syntax errors.
`tar -czf backup.tar.gz data/`	Creates a compressed archive of data/.
`crontab -l`	Displays the current user's scheduled cron jobs.

Before submission: Run every menu option, verify the actual output, and replace the screenshot boxes with screenshots from your own terminal. Confirm the cron job using `crontab -l` and check the log file. The testing table currently marks results as “To be verified” because actual execution results were not provided.